

# Human-Centered Evaluation and Auditing of Language Models

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## ABSTRACT

The recent advancements in Large Language Models (LLMs) have significantly impacted numerous, and will impact more, real-world applications. However, these models also pose significant risks to individuals and society. To mitigate these issues and guide future model development, responsible evaluation and auditing of LLMs are essential. This workshop aims to address the current “evaluation crisis” in LLM research and practice by bringing together HCI and AI researchers and practitioners to rethink LLM evaluation and auditing from a human-centered perspective. The workshop will explore topics around understanding stakeholders’ needs and goals with evaluation and auditing LLMs, establishing human-centered evaluation and auditing methods, developing tools and resources to support these methods, building community and fostering collaboration. By soliciting papers, organizing invited keynote and panel, and facilitating group discussions, this workshop aims to develop a future research agenda for addressing the challenges in LLM evaluation and auditing.

## CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods; HCI theory, concepts and models;** • **Computing methodologies** → **Artificial intelligence.**

## KEYWORDS

Evaluation, Audit, Large Language Models, Generative AI

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## 1 INTRODUCTION

The recent development of Large language models (LLMs), such as GPT-X models, exhibit impressive and versatile capabilities. They are poised to significantly impact the technology landscape, powering numerous applications in various domains. Meanwhile, these models also raise significant risks for individuals and society [46]. They are known to “hallucinate” information—producing plausible but factually incorrect information and potentially contributing to the spread of misinformation. They can generate harmful, sometimes toxic content. They are found to encode biases, which can propagate harmful discrimination, stereotypes, and exclusion [13]. They may also produce inappropriate content or work ineffectively in some contexts or for some user groups more than others.

The key to mitigating the risks of AI technologies is to perform responsible evaluation and auditing. Evaluation and auditing can be performed by different stakeholders [30]: LLM developers must assess whether the model is ready for deployment, and understand when and how the model may fail to enact safe usage policies. Application developers and product managers need the model evaluation results to responsibly decide whether and how to use an LLM. They may also need to perform their own evaluation as they adapt the LLM (i.e., through prompting) or integrate the LLM into their application. End-users and impacted stakeholder groups should be able to assess the model to adapt interactions, provide feedback, and contest the model. Policy-makers and third-party auditors must have means to gauge the model’s limits and risks to have oversight.

However, LLMs are giving rise to an “evaluation crisis” in both AI research and practice. While the AI and NLP communities have traditionally relied on benchmarks to perform automated evaluation for specialized models (e.g., machine translation, summarization, QA models), these benchmarks suffer from many limitations in terms of validity and quality to begin with [16], and they are not equipped to handle the diverse, complex, and uncertain capabilities of LLMs. While initial research efforts in creating LLM-specific meta-benchmarks that can capture multiple model capabilities have

emerged [17, 27], their ability to predict LLMs' utility for different downstream applications and stakeholder groups remains unclear. Researchers have also grappled with other fundamental challenges to the evaluation of LLMs, such as output variability and sensitivity to prompts, and unknown data contamination (i.e., the model could have seen a given evaluation test in its training data).

In practice, some forms of human evaluation such as rating outputs by domain experts or crowd workers are often used to evaluate language models. However, these practices are largely ad-hoc and have been widely criticized for lacking standardization, reproducibility, and validity for assessing model utility in real-world settings [6]. It is also known that practitioners engage in extensive internal auditing (e.g., red teaming [14]) to mitigate the safety concerns and other risks of LLMs, although there is a lack of transparency and shared practices in how such audits are performed [34, 37, 38, 40]. Moreover, the fundamentally socio-technical and contested nature of AI fairness and transparency adds unique challenges in conducting human-centered evaluation and auditing of LLMs [15, 41, 51]. For example, HCI researchers have surfaced challenges around the potential blindspots when the evaluation and auditing of AI systems are only conducted by small groups of technical experts, without meaningful engagements from diverse stakeholders [10, 18, 33, 50].

To incorporate the perspectives and lived experiences of diverse stakeholders in AI evaluation and auditing, there's growing interest among both HCI and AI researchers in directly involving these stakeholders in the evaluation and auditing of LLMs (as well as other AI systems) [3, 19, 22, 35, 36, 39]. In parallel with these research efforts, many major technology companies have begun to experiment with approaches that directly engage users in evaluating and auditing their LLM-powered products and services for problematic behaviors. For instance, OpenAI, inspired by Twitter's "algorithmic bias bounty," [5], initiated a series of feedback contests and red teaming efforts to engage end users in providing feedback on problematic model outputs during their interactions with the ChatGPT [1]. However, AI practitioners often face practical challenges when conducting LLM audits, such as identifying and recruiting the right groups of stakeholders [7, 11], effectively guiding users in auditing LLMs [7, 39], and aggregating the auditing reports from a vast user base [3, 7].

Given these challenges, there is a growing call for rethinking *what* to evaluate and *how* to evaluate & audit LLMs. This workshop aims to bring together HCI and AI researchers and practitioners to contemplate these questions by taking human-centered perspectives, starting by asking: *Who* should be involved in evaluating and auditing LLMs? What are their needs and goals? What are their constraints and challenges? How would they make use of the evaluation and auditing results? Furthermore, we believe it is timely to bring HCI lessons and perspectives about technology evaluation—a core focus of the HCI discipline—into shaping LLM evaluation practices [31]. These perspectives may include a shift away from automatic evaluation to human-subjects evaluation; embracing diverse quantitative and qualitative, empirical and analytical, evaluation methods; considerations of what constitutes "good" or appropriate evaluation methods; best practices for designing evaluation studies and auditing pipelines; tools, processes, and resources (e.g., datasets, statistical tools) to support LLM evaluation practices; regulations and policies that could prioritize marginalized

community members by giving them a greater voice in the LLM audits, and many more.

In summary, this workshop aims to solicit position papers, organize an invited panel, and bring together workshop attendees for group discussions to share research and knowledge about, and develop a future research agenda for, the following topics:

- Identifying current challenges in LLM evaluation and auditing and reasons behind them.
- Understanding the role of various stakeholders in LLM evaluation and auditing.
- Establishing human-centered evaluation and auditing methods of LLMs.
- Developing tools, processes, and resources to support stakeholders to evaluate and audit LLMs.
- Discussing regulatory measures, public policies, or industry standards around evaluating and auditing LLMs.
- Sharing perspectives about the *what*, *why*, and *how* of LLM evaluation and auditing, and opportunities for HCI research and collaboration between the HCI and AI communities.

## 2 ORGANIZING COMMITTEE

This workshop is organized by a team of experienced researchers with diverse research backgrounds in model evaluation and auditing and at different stages of their research careers. Collectively, members of the organizing committee contributed to HCI and AI on various topics, including Explainable AI [28, 30, 52], Responsible AI [7–9, 29, 42–44], Language Technologies [2, 20, 25, 45, 48], Model Evaluation [4, 21, 24, 26, 31, 32, 47, 49], and Algorithm Auditing [7, 10, 12, 22, 23]. Members of the organizing committee have successfully hosted various workshops, tutorials, and panels on the topics of human-centered AI, Algorithm Testing and Auditing, and Evaluation of NLP systems, starting important conversations and community buildings around human-centered AI evaluation and auditing. By integrating expertise from both HCI and AI, as well as our extensive prior experiences around organizing workshops, we aim to bridge the gap between HCI and AI, promote effective discussion, facilitate interdisciplinary collaboration, and galvanize community building.

**Ziang Xiao** is an assistant professor in the Department of Computer Science at Johns Hopkins University. Ziang's work explores the role of language technology in social science discovery, individual information seeking, and evaluation of intelligent systems. His work lies in the intersection of human-computer interaction, natural language processing, and social and personality psychology. Ziang is on the organizing committee and an associate chair for multiple HCI venues (CHI, CSCW, IUI). He will co-organize the third Workshop on Bridging Human-Computer Interaction and Natural Language Processing at NAACL 2024 and co-teach the tutorial on Human-centered Evaluation of Language Technologies at EMNLP 2024.

**Wesley Hanwen Deng** is a Ph.D. student at the Human-Computer Interaction Institute (HCII) at Carnegie Mellon University. Drawing on approaches from human-computer interaction, machine learning, design, and organizational science, Wesley designs and develops tools and processes to support responsible AI practice in

real-world contexts. His current works aim to scaffold effective communication and collaborations between AI developers and everyday users. Before joining CMU, Wesley completed his BA in Computer Science with the Highest Honor at UC Berkeley, where he also received an EECS research honor. Wesley led the efforts to organize a Workshop on “User Engagement in Algorithm Testing and Auditing” at FAccT 2023 and a Workshop on “Supporting User Engagement in Testing, Auditing, and Contesting AI” at CSCW 2023.<sup>1</sup>

**Michelle S. Lam** is a Ph.D. student in Computer Science at Stanford University in the Human-Computer Interaction Group. She builds systems that empower everyday users to actively reshape the design and evaluation of AI systems. Her work on novel methods for human-centered machine learning and algorithm auditing have received awards at top human-computer interaction venues like CHI and CSCW. She is supported by a Stanford Interdisciplinary Graduate Fellowship and has been a Stanford HAI Graduate Fellow, Stanford Technology and Racial Equity Graduate Fellow, Brown Institute for Media Innovation Magic Grantee, and Siebel Scholar. She holds an MS and BS in Computer Science from Stanford University. Michelle helped to co-organize the “Supporting User Engagement in Testing, Auditing, and Contesting AI” at CSCW 2023.

**Motahhare Eslami** is an assistant professor at the School of Computer Science, Human-Computer Interaction Institute (HCII), and Software and Societal Systems Department (S3D), at Carnegie Mellon University. Motahhare’s research goal is to investigate the existing accountability challenges such as auditing in algorithmic systems and to empower the users of algorithmic systems, particularly those who belong to marginalized communities and those whose decisions impact marginalized communities, make transparent, fair, and informed decisions in interaction with algorithmic systems. Motahhare is named one of the 100 Brilliant Women in AI Ethics. Her work has been recognized with a Google Ph.D. Fellowship, Best Paper and Honorable Mention Awards, as well as award for contribution to inclusion and diversity at top-tier ACM conferences, and has been covered in mainstream media such as Time, The Washington Post, Huffington Post, the BBC, Fortune, Quartz, and WIRED. Motahhare is named one of the 100 Brilliant Women in AI Ethics, and her research is supported by NSF (Fairness in AI, Future of Work, and AI Institute), CMU’s Block Center for Technology and Society, Amazon, Google, Meta, and Cisco.

**Juho Kim** is an Associate Professor in the School of Computing at KAIST, affiliate faculty in the Kim Jaechul Graduate School of AI at KAIST, and a director of KIXLAB (the KAIST Interaction Lab). His research in human-computer interaction and human-AI interaction focuses on building interactive and intelligent systems that support interaction at scale, aiming to improve the ways people learn, collaborate, discuss, make decisions, and take action online. He earned his Ph.D. from MIT, M.S. from Stanford University, and B.S. from Seoul National University. In 2015-2016, he was a Visiting Assistant Professor and a Brown Fellow at Stanford University. He is a recipient of a KIISE/IEEE-CS Young Computer Researcher Award, KAIST’s Songam Distinguished Research Award, Grand Prize in Creative Teaching, Q-Day Creative Education Award, and Excellence in Teaching Award, as well as 14 paper awards from ACM CHI, ACM CSCW, ACM Learning at Scale, ACM IUI, ACM

DIS, and AAAI HCOMP. He recently spent his sabbatical year as a chief scientist at Ringle Inc. to transfer his research on AI-powered analysis and diagnosis of English learners’ proficiency into a real product. He gave a keynote at NeurIPS 2022 titled “Interaction-Centric AI”.

**Mina Lee** is a postdoctoral researcher in the Computational Social Science group at Microsoft Research as well as an incoming assistant professor in Computer Science at the University of Chicago in 2024. Her research aims to (i) design and evaluate language models to enhance our productivity and creativity and (ii) understand how these models change the way we write. She has built various writing assistants, including an autocomplete system, a contextual thesaurus system, and a creative story writing system, as well as evaluated language models based on their ability to interact with humans and augment human capabilities. She was named one of MIT Technology Review’s Korean Innovators under 35 in 2022. She co-founded and organized the Workshop on Intelligent and Interactive Writing Assistants (In2Writing) for the past two years at ACL 2022 and CHI 2023.<sup>2</sup>

**Q. Vera Liao** is a Principal Researcher at Microsoft Research Montreal, where she is part of the FATE (Fairness, Accountability, Transparency, and Ethics) group. Her current research interests are in human-AI interaction, explainable AI, and responsible AI, with an overarching goal of bridging emerging AI technologies and human-centered research and design practices. Prior to joining MSR, she worked at IBM Research, and studied at the University of Illinois at Urbana-Champaign and Tsinghua University. She has co-organized many workshops at CHI, IUI and NeurIPS, including CHI 2021 and 2022 Workshops on Human-Centered Explainable AI, and CHI 2023 Workshop on GenAI and HCI.

### 3 PLANS TO PUBLISH WORKSHOP PROCEEDINGS

To find suitable attendees (approximately 50-60 participants), we will ask participants to submit a position paper or short paper with preliminary findings (up to four pages) on human-centered evaluation and auditing of LLMs. Hence, the workshop organizers will put out a call to solicit papers and form a program committee to review and carefully select the incoming submissions. We plan to provide at least two reviews for each submission to assess the quality and relevance of participants’ content. The authors of selected submissions will be invited to participate in our workshop. This way, we will control the number of participants at our workshop to avoid physical space shortages.

The primary aim of this workshop is to build a new research community on the topic to share future research agendas and promote collaboration among researchers from various backgrounds and viewpoints. All accepted papers from attendees will be published on the workshop’s website to guarantee widespread distribution of the opinions and findings. Furthermore, the accepted papers will be featured in the workshop proceedings, which we hope to publish on *ceur-ws*.<sup>3</sup> Finally, we will actively encourage all authors to submit extended versions of their papers for potential inclusion in a special issue in a well-recognized HCI journal.

<sup>1</sup><https://cscw-user-ai-auditing.github.io/>

<sup>2</sup><https://in2writing.glitch.me/>

<sup>3</sup><https://ceur-ws.org/>

Ideas and discussion topics will be collected throughout the session in a shared document, both during and after the workshop. Attendees will also be invited to participate in future joint projects stemming from the discussion and presentations at the workshop.

## 4 IN-PERSON AND HYBRID

We acknowledge the numerous challenges and concerns that attendees may face, including visa complications, financial limitations, and personal obligations, which could potentially prevent them from participating in an in-person workshop. Since the main goal for this workshop is community building, we believe it is crucial to make our workshop open and inclusive for everyone. Therefore, we plan to host the workshop in a hybrid form. We are dedicated to supporting the CHI's initiative about fostering an inclusive environment for both in-person and hybrid attendees.<sup>4</sup> We will facilitate synchronous discussions for both in-person attendees with dedicated cameras and remote attendees on Zoom. We will utilize online platforms like Miro to create a coherent experience for both in-person and remote attendees during the discussion sessions. All attendees will use the same Slack channel to share workshop materials and connect with each other. All keynote, panels, and presentations will be live for remote participants with live captioning.

## 5 ASYNCHRONOUS ENGAGEMENT

We will utilize email lists, social media and workshop websites to encourage participation and share key information. The website will function as a central place for distributing information, such as the workshop description and proposal, a call for participation, the workshop schedule, and information about the keynote speaker and panelists.

We will create a dedicated Slack group to facilitate asynchronous engagement. Before the workshop, all registered attendees will be invited to the Slack group for pre-workshop activities. We will survey attendees to ask about their interests in this topic and ideal takeaways. During the workshop, the Slack channel will be used to disseminate important information and share workshop materials. After the workshop, the Slack group will be actively maintained by the organizers for attendees to further engage with each other, discuss related issues, share research outcomes, and incubate collaborations.

We will strongly encourage attendees to create a 5-minute video presentation for their accepted workshop paper, which will be shown on the workshop's website with captions. We will also record the keynote and panel talk during the workshop. Attendees can access the video recordings asynchronously on Slack.

## 6 WORKSHOP STRUCTURE

We design a workshop structure for the goal of building community, exchanging knowledge and co-creating new research agendas for this emerging topic. We plan for a one-day, in-person/hybrid workshop structured in a series of presentations, activities, and guided discussions. We are aiming for approximately 50-60 attendees. Given we will use our pre-workshop survey to guide the exact topics for the discussion section, our tentative schedule is below.

- (1) **Welcome (15 mins):** Opening talk by one of the organizers, with brief introductions from organizers, highlighting the workshop's goals and going over plans for the day.
- (2) **Opening Keynote (45 mins):** A distinguished keynote speaker will deliver a 30-minute opening keynote with their insights and perspectives on human-centered evaluation and auditing of LLMs. Following the presentation, there will be a 15-minute Q&A session, providing an opportunity for attendees to engage with the speaker and ask questions related to the topic.
- (3) **Coffee Break (15 mins)**
- (4) **Author Presentation I (60 mins):** Authors of accepted papers will give a 5-minute lightning talk about their paper. At the end, we will have 15 minutes to invite authors as a panel to answer the audience's questions.
- (5) **Discussion Session I - Challenges Identification (60 mins):** Attendees will split into smaller groups and discuss the main challenges and goals in LLM evaluation and auditing and their impact on stakeholders in downstream applications. Each group will identify real-world challenges for LLM evaluation and auditing and discuss how current evaluation and auditing practices fell short. We will have designated organizers to facilitate an online discussion session for remote attendees.
- (6) **Lunch Break (90 mins)**
- (7) **Panel (60 mins):** Our panel will consist of senior researchers and practitioners working on human-centered evaluation and auditing of LLMs. The panel will kick off the workshop by discussing different aspects of LLM evaluation and auditing, such as unique challenges LLM stakeholders face in understanding model capability, novel evaluation and audit methods for LLMs, and bridging HCI and AI communities on shared challenges. This panel will set a solid foundation for the upcoming activities.
- (8) **Author Presentation II (60 mins):** Authors of accepted papers will give a 5-minute lightning talk about their paper. At the end, we will have 15 minutes to invite authors as a panel to answer the audience's questions.
- (9) **Discussion Session II - Methods and Tooling (60 mins):** In this session, attendees will explore various design techniques and methodologies that can be employed in LLM evaluation and auditing. Groups will brainstorm and discuss creative approaches to improve the evaluation process and auditing practice. Discussions will focus on identifying appropriate methods for evaluation goals and exploring the trade-offs between various methods. We will have designated organizers to facilitate an online discussion session for remote attendees.
- (10) **End Note (30 mins):** The organizers will end the workshop by synthesizing the main points from two discussions in previous discussion sessions. Attendees and organizers will reflect together to develop tangible future research topics.

The workshop will alternate between lightning talks, Q&A, group activities, and plenary discussions. The current discussion topics are tentative, and we will change them based on the attendees' interests in the pre-workshop survey. The lightning talks order will also be

<sup>4</sup><https://chi2024.acm.org/for-attendees/accessibility-faq/>

arranged based on the discussion topic. We will facilitate different groups of attendees interaction to provide attendees with ample opportunities to exchange ideas and form new collaborations.

## 7 POST-WORKSHOP PLANS

The expected workshop outcomes include:

- An active growing community of researchers who are interested in working on human-centered evaluation and auditing for LLMs to exchange ideas and form collaborations. We plan to create email list and a Slack channel to further engage with participants post-workshop.
- An article for the ACM Interactions magazine or a blog post to summarize the workshop outcome and future research landscape for both AI and HCI audiences.
- A workshop proceedings with accepted papers published on *ceur-ws*.<sup>5</sup>
- An invitation to authors of selected accepted papers for a special issue at a well-regarded HCI journal (e.g. TiiS, TOCHI).

## 8 ACCESSIBILITY

We will survey attendees before the workshop about their accessibility needs and attending constraints. We will provide essential materials and tools such as large-print handouts, assistive listening devices, and accessible seating to ensure an inclusive design workshop experience for all attendees. We will strongly encourage all attendees to adhere to the ACM's Accessible Submission Guide<sup>6</sup> when preparing their manuscript and encourage them to perform an accessibility check for the final camera-ready version. To guarantee an inclusive experience for in-person and hybrid attendees, we will ensure automatic transcription services are available for panel and lighting talks. Additionally, to promote accessibility and maintain thorough documentation, assigned note-takers will record crucial points in a shared Google Doc, enabling attendees to revisit discussions and input after the workshop. With conference organizers' support, we will incorporate diverse engagement methods, including video, audio, and text, to accommodate attendees with varied visual, hearing, speech, and cognitive abilities.

## 9 CALL FOR PARTICIPATION

As Large Language Models (LLMs) continue to advance, addressing the "evaluation crisis" and exploring human-centered evaluation and auditing approaches become increasingly important. Ensuring that these models are effective, ethical, and socially responsible is crucial for their widespread adoption and integration into many real-world applications. We invite authors from all disciplines to submit papers to the HEAL@CHI'24: Human-centered Evaluation and Auditing of Language Models<sup>7</sup>.

This workshop aims to foster discussions on understanding real-world stakeholders' needs and goals, establishing human-centered evaluation and auditing methods for LLMs, and developing tools, processes, and resources to support stakeholders in evaluating and auditing LLMs.

Papers can address topics, but not limited to, such as:

- Empirical understanding of stakeholders' needs and goals of LLM evaluation and auditing
- Human-centered evaluation and auditing methods for LLMs
- Tools, processes, and guidelines for LLM evaluation and auditing
- Discussion of regulatory measures and public policies for LLM auditing
- Ethics in LLM evaluation and auditing

### Submission Requirements:

Position papers should be up to 4 pages in the CHI Publication Format, excluding references. Submit papers to [Open Review Link].<sup>8</sup>

Submissions will go through a jury review process and be selected based on quality. Accepted papers will be featured on our website and presented during the workshop, and at least one author must attend the workshop.

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<sup>8</sup>We plan to use OpenReview to manage the paper submission process.

<sup>5</sup><https://ceur-ws.org/>

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